

Name : \_\_\_\_\_  
Class : 12 MT \_\_\_\_\_

CHERRYBROOK TECHNOLOGY HIGH SCHOOL

1999 AP3

YEAR 12 HALF YEARLY HSC

**MATHEMATICS**  
**3/4 UNIT (COMMON)**

*Time allowed - 2 HOURS*  
*(plus 5 minutes' reading time)*

**DIRECTIONS TO CANDIDATES:**

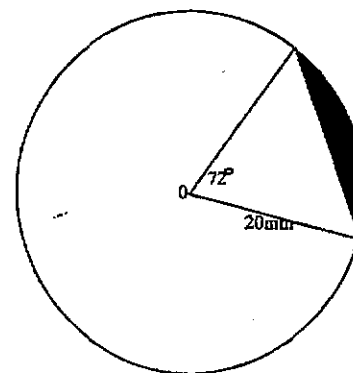
- \* Attempt ALL questions.
- \* All questions are of equal value
- \* All necessary working should be shown in every question.  
Full marks may not be awarded for careless or badly arranged work.
- \* Standard Integrals are provided. Approved calculators may be used.
- \* Each question attempted is to be returned on a new page clearly marked Question 1, Question 2, etc on the top of the page.

\*Each page must show your class and your name.

**QUESTION 1**

**Marks**

- (a) Expand  $(2x - y)^5$  2
- (b) (i) Write down the expansion of  $\cos(\alpha - \beta)$ . 3
- (ii) Find the exact values of  $\cos 45^\circ$  and  $\cos 30^\circ$ .
- (iii) Hence find the exact value of  $\cos 15^\circ$ .
- (c) (i) Convert  $72^\circ$  to radians, giving your answer in terms of  $\pi$ . 3
- (ii) Hence or otherwise, find the shaded area below correct to 3 significant figures.



- (d) Solve  $\sin 2x = \sqrt{3} \cos 2x, 0 \leq x \leq 2\pi$ . 2
- (e) Differentiate with respect to  $x$  2
- (i)  $\sqrt[3]{4x - 1}$
- (ii)  $\frac{x}{\cot x}$

QUESTION 2

BEGIN A NEW PAGE

Marks

(a) Given  $\int_0^3 f(t) dt = 6$ , evaluate

4

(i)  $\int_0^1 f(t) dt + \int_1^3 (f(t) + 1) dt$

(ii)  $\int_3^0 (f(t) + t) dt$

(b) Find

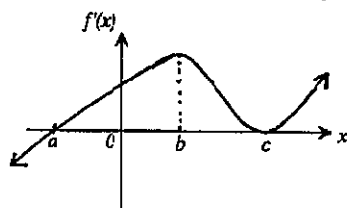
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(i)  $\int \frac{x^4 + 2x^3 + 3}{x^2} dx$

(ii)  $\int \frac{dt}{(3-t)^2}$

(c) The gradient function of  $y = f(x)$  is graphed below.

6



(i) Copy this diagram onto your answer sheet.

(ii) On the same diagram, sketch and label a possible graph of  $y = f''(x)$ .

(iii) State the domain where  $y = f(x)$  is concave down.

(iv) Find the  $x$  values of any points of inflection.

(v) Find any stationary points and determine their nature.

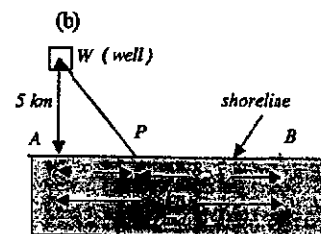
QUESTION 3

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Marks

(a) Find the equation of any asymptotes of the curve  $y = \frac{x^2 + x + 1}{x}$ .

2



An offshore oil well is located at a point  $W$ , which is 5 km from the closest shorepoint  $A$  on a straight shoreline.

The oil is to be piped to a shorepoint  $B$  that is 8 km from  $A$  by piping it on a straight line under water from  $W$  to some shorepoint  $P$  between  $A$  and  $B$  and then on to  $B$  via a pipe along the shoreline.

If the cost of laying the pipe is \$125 000 per km under water and \$75 000 per km over land.

Let  $x$  km be the distance between  $A$  and  $P$  and  $C$  (in thousands of dollars) be the cost for the entire pipeline.

(i) Show that the cost is given by  $C = 125\sqrt{x^2 + 25} + 75(8 - x)$

3

(ii) Find the domain for  $x$

1

(iii) Find where the point  $P$  should be located to minimise the cost of laying the pipe?

6

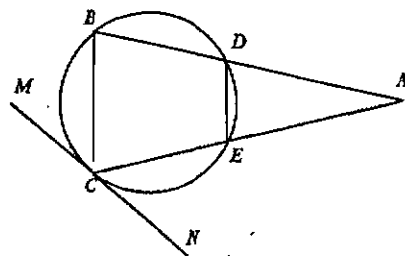
QUESTION 4

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Marks

(a)

6



$ABC$  is a triangle in which  $AB = AC$ . A circle through  $B$  and  $C$  cuts  $AB$  at  $D$  and  $AC$  at  $E$ .  $MCN$  is the tangent at  $C$  to the circle through  $B, C, E, D$ .

(i) Copy the diagram onto your answer sheet.

(ii) Show that  $DE \parallel BC$ .

(iii) Show that  $\angle ACN = \angle BCD$ .

(b)  $P(2at, at^2)$  is a variable point on the parabola  $x^2 = 4ay$ , whose focus is  $S$ .

6

$Q(x, y)$  divides the interval from  $P$  to  $S$  in the ratio  $t^2 : 1$ .

(i) Find  $x$  and  $y$  in terms of  $a$  and  $t$ .

(ii) Verify that  $\frac{y}{x} = t$ .

(iii) Prove that as  $P$  moves on the parabola,  $Q$  moves on a circle, and state its centre and radius.

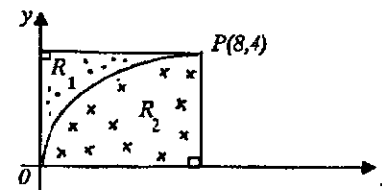
QUESTION 5

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Marks

(a)

7



$OP$  is an arc of the curve  $y^3 = x^2$ .

Calculate the volume of the solids generated when

(i) Region  $R_1$  revolves around the  $y$ -axis

(ii) Region  $R_2$  revolves around the  $x$ -axis

(iii) Region  $R_2$  revolves about the  $y$ -axis.

(b) (i) Express  $\sin x - \cos x$  in the form  $A \sin(x - \alpha)$  with  $A > 0$  and  $0 < \alpha < \frac{\pi}{2}$ .

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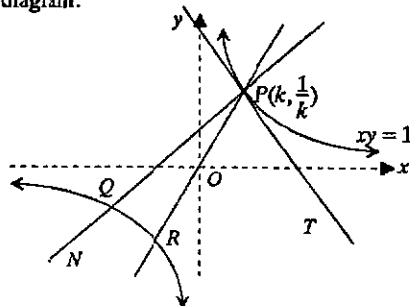
(ii) Determine  $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{x - \frac{\pi}{4}}$ .

QUESTION 6

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Marks

$P(k, \frac{1}{k})$  is a point on the curve  $xy = 1$  where  $k$  is a real number,  $k \neq 0$ .  
 $PT$  is the tangent to the curve at  $P$  and  $PN$  is the normal at  $P$ .  
 $POR$  is the line passing through  $P$ , the Origin  $O$  and  $R$  as shown on the diagram.



- (a) Find the equation of the line passing through  $O$  and  $P$ . 1
- (b) The line in part (a) intersects the curve again at  $R$ . Find the coordinates of  $R$ . 2
- (c) Show that the equation of the tangent at  $P$  is given by:  

$$x + k^2y = 2k.$$
 2
- (d) Find the equation of the normal line at  $P$ . 2
- (e) Show that when the normal intersects the curve again at  $Q$ , the equation formed to solve is the quadratic equation given by:  

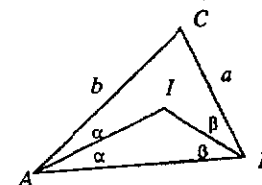
$$k^3x^2 - (k^4 - 1)x - k = 0.$$
 Hence find the coordinates of point  $Q$  3
- (f) Show that:  $QR \perp PR$ . 2

QUESTION 7

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Marks

- (a) Given the polynomial function  $P(x) = x^3 - 2x^2 - 6x + 4$ , when  $P(x) = 0$ ,  $P(x)$  has one rational root and two irrational roots.
  - (i) Find the rational root of  $P(x) = 0$ . 1
  - (ii) Without finding the irrational roots of  $P(x)$ , show that one of the irrational roots of this equation lies between  $x = 3$  and  $x = 4$ . 1
  - (iii) Using  $x = 3.5$  as a first approximation, apply Newton's Method once to find a better approximation to the root, to 2 decimal places. 3
  - (iv) Sketch  $P(x) = x^3 - 2x^2 - 6x + 4$ . 1
  - (v) Explain why  $x = 2$  would not be a good approximation to use when solving  $P(x) = 0$  using Newton's Method. 1
  - (vi) Find the area bounded by the curve,  $x = -3$ ,  $x = -2$  and the  $x$ -axis. 2
- (b)  $IA$  and  $IB$  bisect angles  $CAB$  and  $CBA$  as shown in the diagram below. 3



Prove that  $\frac{IB}{IA} = \frac{a \cos \beta}{b \cos \alpha}.$

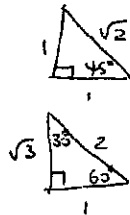
Question 1

$$\begin{aligned}
 (a) (2x-y)^5 &= 1(2x)^5(-y)^0 + 5(2x)^4(-y)^1 + 10(2x)^3(-y)^2 + 10(2x)^2(-y)^3 + 5(2x)^1(-y)^4 + (2x)^0(-y)^5 \\
 &= 32x^5 - 80x^4y + 80x^3y^2 - 40x^2y^3 + 10xy^4 - y^5
 \end{aligned}$$

①

$$(b) (i) \cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta \quad ①$$

$$\begin{aligned}
 (ii) \cos 45^\circ &= \frac{1}{\sqrt{2}} \\
 \cos 30^\circ &= \frac{\sqrt{3}}{2}
 \end{aligned}$$

① need both

$$\begin{aligned}
 (iii) \cos 15^\circ &= \cos(45^\circ - 30^\circ) \\
 &= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ \\
 &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \\
 &= \frac{\sqrt{3} + 1}{2\sqrt{2}} \quad ① \\
 &= \frac{\sqrt{6} + \sqrt{2}}{4}
 \end{aligned}$$

3

$$(c) (i) 1^\circ = \frac{\pi}{180}$$

$$72^\circ = \frac{72\pi}{180}$$

$$= \frac{8\pi}{20}$$

$$= \frac{2\pi}{5} \quad ①$$

$$\begin{aligned}
 (ii) \text{Area of minor segment} &= \frac{1}{2} r^2 (\theta - \sin \theta) \\
 &= \frac{1}{2} \cdot 20^2 \cdot \left( \frac{2\pi}{5} - \sin \frac{2\pi}{5} \right) \quad ① \\
 &= 61.11610 \dots \\
 &= 61.1 \text{ mm}^2 \text{ (3 sig. fig.)} \quad ①
 \end{aligned}$$

3

$$(d) \sin 2x = \sqrt{3} \cos 2x$$

$$\frac{\sin 2x}{\cos 2x} = \sqrt{3}$$

$$\tan 2x = \sqrt{3}$$

$$2x = \frac{\pi}{3}, \pi + \frac{\pi}{3}, 2\pi + \frac{\pi}{3}, 3\pi + \frac{\pi}{3}, \dots \quad ①$$

$$x = \frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6}, \frac{19\pi}{6}, \dots \quad ①$$

2

$$\begin{aligned}
 (e) (i) \frac{d}{dx} \sqrt[3]{4x-1} &= \frac{d}{dx} (4x-1)^{\frac{1}{3}} \\
 &= \frac{1}{3} \cdot (4x-1)^{-\frac{2}{3}} \cdot 4 \\
 &= \frac{4}{3} (4x-1)^{-\frac{2}{3}} \text{ or } \frac{4}{3 \sqrt[3]{4x-1}^2} \quad ①
 \end{aligned}$$

$$\begin{aligned}
 (ii) \frac{d}{dx} \left( \frac{x}{\cot x} \right) &= \frac{d}{dx} x \tan x \\
 &= x \cdot \sec^2 x + \tan x \cdot 1 \\
 &= x \sec^2 x + \tan x \quad ①
 \end{aligned}$$

2

Question 2

$$\begin{aligned} (a) (i) &= \int_0^1 f(t) dt + \int_1^3 f(t) dt + \int_3^4 1 dt \\ &= \int_0^3 f(t) dt + [t]_1^3 \\ &= 6 + 3 - 1 \\ &= 8 \end{aligned} \quad \textcircled{1}$$

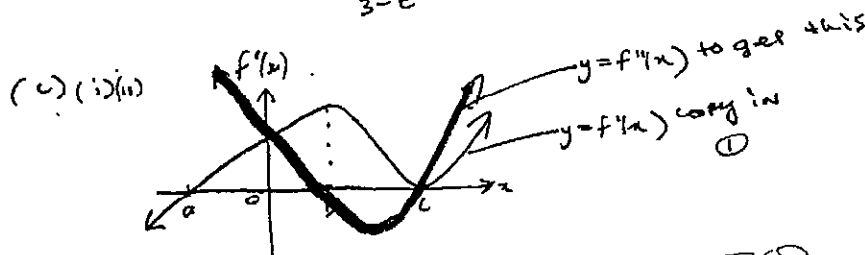
$$\begin{aligned} (ii) &= \int_3^0 f(t) dt + \int_3^0 t dt \\ &= - \int_0^3 f(t) dt + \left( \frac{t^2}{2} \right)_3^0 \\ &= -6 + 0 - \frac{9}{2} \\ &= -10\frac{1}{2} \end{aligned} \quad \textcircled{1}$$

4

$$\begin{aligned} (b) (i) & \int \frac{x^4 + 2x^3 + 3}{x^2} dx \\ &= \int (x^2 + 2x + 3x^{-2}) dx \\ &= \frac{x^3}{3} + x^2 - 3x^{-1} + C \\ &= \frac{x^3}{3} + x^2 - \frac{3}{x} + C \end{aligned} \quad \textcircled{1}$$

$$\begin{aligned} (ii) & \int \frac{dt}{(3-t)^2} = \int (3-t)^{-2} dt \\ &= \frac{(3-t)^{-1}}{-1 \times -1} \\ &= \frac{1}{3-t} + C \end{aligned} \quad \textcircled{1}$$

2



- (i) }
- (ii) }
- (iii) |
- (iv) 1+1

(iii)  $b < x < c$

(iv)  $x=b$

| $x$     | $b^-$ | $b$ | $b^+$ |
|---------|-------|-----|-------|
| $f'(x)$ | +     | 0   | -     |

There is a change in concavity at  $x=b$

| $x$     | $c^-$ | $c$ | $c^+$ |
|---------|-------|-----|-------|
| $f'(x)$ | -     | 0   | +     |

There is a change in concavity at  $x=c$

$\therefore$  Point of inflection at  $x=b, x=c$

(v) 1+1

(i)  $f'(a) = 0$

$f''(a) > 0$

$\therefore x=a$  is a rel. min. turning pt

Tests

$f'(x) > 0$

| $x$     | $c^-$ | $c$ | $c^+$ |
|---------|-------|-----|-------|
| $f'(x)$ | +     | 0   | +     |
| Slope   | —     | —   | —     |

$\therefore x=c$  is a stationary point of inflection  
 H.P.O.I

6

### Question 3

$$(a) y = \frac{x^2 + x + 1}{x}$$

$$y = x + 1 + \frac{1}{x}$$

vertical asymptote  $x=0$

diagonal asymptote  $y=x+1$

2

$$(b) WP^2 = 25 + x^2$$

$$WP = \sqrt{25 + x^2}$$

$$\text{Cost} = 125000 \times WP + 75000 \times PB \quad (1)$$

$$C = 125 \sqrt{25 + x^2} + 75(8 - x) \quad (C \text{ is in thousands of dollars})$$

3

$$(ii) x^2 + 25 > 0 \quad \text{and} \quad 8 - x \geq 0 \quad \text{and} \quad x \geq 0$$

$$x^2 > -25$$

$$\therefore x \in \mathbb{R}$$

$$-x \geq -8$$

$$x \leq 8$$

$$\therefore 0 \leq x \leq 8$$

1

1

$$(iii) \frac{dC}{dx} = 125 \cdot \frac{1}{2} (25 + x^2)^{-\frac{1}{2}} \cdot 2x - 75$$

$$= \frac{125x}{\sqrt{25 + x^2}} - 75$$

$$\frac{dC}{dx} = 0 \quad \text{for max/min}$$

$$\frac{125x}{\sqrt{25 + x^2}} - 75 = 0$$

$$125x = 75\sqrt{25 + x^2}$$

$$5x = 3\sqrt{25 + x^2}$$

Square both sides

$$25x^2 = 9(25 + x^2)$$

$$25x^2 = 225 + 9x^2$$

$$16x^2 = 225$$

$$x^2 = \frac{225}{16}$$

$$x = \pm \frac{15}{4}$$

Since  $0 \leq x \leq 8$

$$x = \frac{15}{4}$$

(1) This allocated at end of question after testing

|                 | 3              | 4              |
|-----------------|----------------|----------------|
| $x$             | $\frac{15}{4}$ | $\frac{15}{4}$ |
| $\frac{dC}{dx}$ | -              | +              |
| Sign            | -              | +              |

$$\therefore \text{rel. min when } x = \frac{15}{4}$$

test end points of  $0 \leq x \leq 8$

$$x = 0$$

$$\text{Cost} = (5 \times 125 + 8 \times 75) \times 1000$$

$$= \$1225000$$

$$x = 8$$

$$WP = \sqrt{8^2 + 25}$$

$$PB = 0$$

$$\therefore \text{Cost} = (\sqrt{89} \times 125) \times 1000$$

$$= \$1179247.642 \dots$$

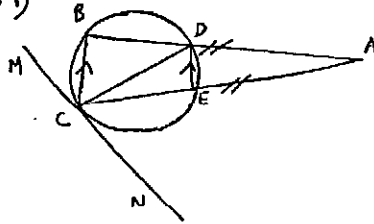
$$= \$1179247.64$$

$\therefore$  The point P should be located  $\frac{15}{4} = 3.75 = 3.75 \text{ km}$

1

### Question 4

(a) i)



ii)  $\triangle ABC$  is isosceles ( $AB = AC$ )

Let  $\angle ABC = x$

$\therefore \angle ABC = \angle BCA = x$  (equal base  $\angle$ 's, isos  $\Delta$ ) (1)

$$\angle BDE = 180^\circ - \angle ACB \quad (\text{opp } \angle \text{ s yz due to } \parallel \text{ are supp})$$

$$= 180^\circ - \alpha \quad (1)$$
$$\angle CAD + \angle BDE = 180^\circ \quad (1)$$

and these are wintering notes ①

∴ BC || DE

iii)  $\angle ACN = \angle CDE$  (angle in the alternate segment theorem) (1)

$$\angle CDE = \angle BCD \quad (\text{alt- } \angle\text{'s } \Rightarrow BC \parallel DE) \quad (1)$$
$$\therefore \angle ACN = \angle BCD$$

(b)

A graph of a parabola  $x^2 = 4ay$  is shown on a Cartesian coordinate system. The parabola opens upwards with its vertex at the origin  $(0,0)$ . A point  $P(2at, at^2)$  is marked on the parabola in the first quadrant. A line segment connects the origin  $(0,0)$  to point  $P$ . A tangent line is drawn from a point  $(0,p)$  on the y-axis to point  $P$ . The line segment from  $(0,p)$  to  $P$  is perpendicular to the tangent line at  $P$ . The distance from the origin to  $P$  is labeled as  $S$ .

$$\begin{aligned} (i) \quad \bar{Q} &= \left( \frac{nx_1 + m\bar{x}_2}{n+m}, \frac{ny_1 + my_2}{n+m} \right) \\ &= \left( \frac{1 \times 2at^2 + t^2 \times 0}{t^2 + 1}, \frac{1 \times at^2 + t^2 \times a}{t^2 + 1} \right) \\ &= \left( \frac{2at^2}{t^2 + 1}, \frac{2at^2}{t^2 + 1} \right) \end{aligned}$$

$$(11) \frac{y}{x} = \frac{\frac{2x+2}{t^2+1}}{\frac{2x}{t^2+1}} = \frac{2x+2}{2x}$$

$$\therefore \frac{y}{x} = b \quad (1)$$

$$\begin{aligned}(m) \quad x &= \frac{2xy}{x^2+1} \\ &= \frac{2a\left(\frac{y}{x}\right)}{\left(\frac{y}{x}\right)^2+1} \\ &= \frac{\frac{2ay}{x}}{\frac{y^2+x^2}{x^2}}\end{aligned}$$

$$\therefore x = \frac{2xy}{x^2 + y^2} \quad (1)$$

$$3x^2 + y^2 = 2xy$$

$$x^2 + y^2 - 2ay = 0$$

$$x^2 + (y-a)^2 = a^2 \quad (1)$$

circle centre  $(0, -2)$

admission a unit

30

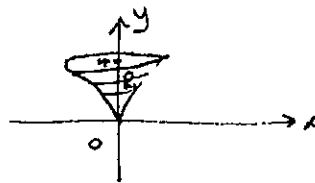
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# Question 5

$$(a) (i) \text{ Volume} = \pi \int_0^4 y^3 dx$$

$$= \pi \left[ \frac{y^4}{4} \right]_0^4 \quad (1)$$

$$= \pi \left[ \frac{4^4}{4} - 0 \right]$$



$$= 64\pi \text{ cubic units} \quad (1)$$

$$(ii) \text{ Volume} = \pi \int_0^8 x^{4/3} dx \quad (1)$$

$$= \pi \left[ \frac{3x^{7/3}}{7} \right]_0^8 \quad (1)$$

$$= \pi \left[ \frac{3}{7} 8^{7/3} - 0 \right]$$

$$= \pi \cdot \frac{3}{7} \cdot 2^7$$

$$= \frac{384}{7} \pi \text{ cubic units} \quad (1)$$

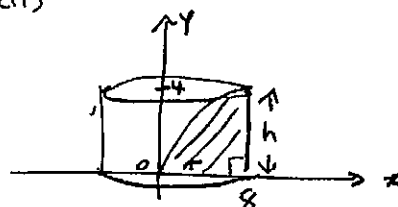
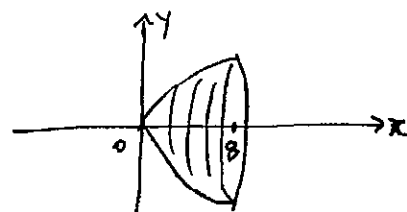
$$(iii) \text{ Volume} = \pi \times 8^2 \times 4 - 64\pi \quad (1) \quad \leftarrow \text{Qs (i)}$$

$$= 192\pi \text{ cubic units} \quad (1)$$

$$y^3 = x^2$$

$$y = x^{2/3}$$

$$y^2 = x^{4/3}$$



$$(b) (i) \sin x - \cos x = A \sin(x - \alpha)$$

$$= A \sin x \cos \alpha - A \cos x \sin \alpha \quad (1)$$

$$\therefore A \cos \alpha = 1 \quad \dots \dots \dots (1)$$

$$A \sin \alpha = 1 \quad \dots \dots \dots (2)$$

$$(1)^2 + (2)^2$$

$$A^2 \sin^2 \alpha + A^2 \cos^2 \alpha = 1 + 1$$

$$A^2 (\sin^2 \alpha + \cos^2 \alpha) = 2$$

$$A^2 = 2$$

$$A = \sqrt{2} \quad (1)$$

$$\text{From (1), (2)} \quad \left. \begin{array}{l} \cos \alpha = \frac{1}{\sqrt{2}} \\ \sin \alpha = \frac{1}{\sqrt{2}} \end{array} \right\} \text{quad I, } \triangle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 1$$

$$\therefore \alpha = \frac{\pi}{4}$$

$$\sin x - \cos x = \sqrt{2} \sin \left( x - \frac{\pi}{4} \right) \quad (1)$$

$$(ii) \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{x - \frac{\pi}{4}} = \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sqrt{2} \sin \left( x - \frac{\pi}{4} \right)}{x - \frac{\pi}{4}} \quad (1)$$

$$= \sqrt{2} \quad (1)$$

7

5

# QUESTIONS

$$(a) m_{op} = \frac{\frac{1}{k} - 0}{k - 0}$$

$$= \frac{1}{k^2}$$

eqn of is.

$$y = \frac{1}{k^2}x$$

$$x - k^2y = 0$$

①

1

(b) Solve simultaneously  $y = \frac{1}{k^2}x$

$$\text{and } y = \frac{1}{x}$$

$$\therefore \frac{1}{k^2}x = \frac{1}{x}$$

①

$$x^2 = k^2$$

$$x = k \text{ or } x = -k$$

Since P is  $(k, \frac{1}{k})$

$$R \text{ is } (-k, -\frac{1}{k})$$

①

2

$$(c) y = x^{-1}$$

$$\frac{dy}{dx} = -x^{-2}$$

$$= -\frac{1}{x^2}$$

$$\text{at } x = k, \frac{dy}{dx} = -\frac{1}{k^2}$$

$$\therefore \text{gradient of tangent} = -\frac{1}{k^2}$$

①

$$\text{equation of tangent is } y - \frac{1}{k} = -\frac{1}{k^2}(x - k)$$

$$k^2y - k = -x + k$$

$$x + k^2y = 2k \text{ as required}$$

①

$$(d) \text{Slope of tangent} = -\frac{1}{k^2}$$

$$\therefore \text{slope of normal} = k^2$$

$$(m_1 m_2 = -1)$$

①

$$\text{Equation of normal } y - \frac{1}{k} = k^2(x - k)$$

$$ky - 1 = k^3x - k^4$$

$$k^3x - ky = k^4 - 1$$

①

$$y = k^2x + \frac{1}{k} - k^3$$

2

(e) Solve simultaneously

$$y = \frac{1}{x} \text{ and}$$

eqn ①

$$k^3x - ky = k^4 - 1$$

eqn ②

Sub eqn ① into eqn ②

$$k^3x - k\left(\frac{1}{x}\right) = k^4 - 1$$

$$k^3x^2 - k = (k^4 - 1)x$$

$$k^3x^2 - (k^4 - 1)x - k = 0$$

}

①

Since  $P(k, \frac{1}{k})$  lies on the normal

$x=k$  is a root of the equation

or use  $\alpha\beta = \frac{k^4}{k^3}$

method ①

$$\begin{array}{r} k^3x + 1 \\ x-k \overline{) K^3x^2 - (K^4-1)x - K} \\ \underline{K^3x - K^4x} \phantom{- K} \\ x - k \\ \underline{x - k} \\ 0 \end{array}$$

$\therefore (x-k)(k^3x+1)=0$

$x=k$  or  $x = -\frac{1}{k^3}$

Q is  $(-\frac{1}{k^3}, -k^3)$

② or

form

method ②

product of roots

$$\alpha\beta = -\frac{k}{k^3}$$

$$= -\frac{1}{k^2}$$

since  $\alpha=k$  is one root

$$k\beta = -\frac{1}{k^2}$$

$$\beta = -\frac{1}{k^3}$$

$\therefore Q$  is  $(-\frac{1}{k^3}, -k^3)$

1 method  
1 answer

②

③

(f) Slope QR =  $\frac{-\frac{1}{k} - (-k^3)}{-k - (-\frac{1}{k^3})}$

$$= \frac{(k^4 - 1)}{\frac{-k^4 + 1}{k^3}}$$

$$= -k^2$$

①

Slope PR =  $\frac{1}{k^2}$

Slope QR  $\times$  Slope PR =  $-k^2 \times \frac{1}{k^2}$

$$= -1$$

$\therefore QR \perp PR$

①

②